

Glengarry Pl upper deck

· 2 · Fulton rail · new

caissons and 8x8 DF posts

166 Glengarry Pl, Castle Rock · Vintage Coastline, Dark Hickory picture frame and breakers · SYP @ 12" · 6x12 DF beams · Cortex

CHECK OR ACH	FINANCED	MONTHLY	SQUARE FEET	FOOTINGS	RAIL
\$140,327	\$150,889	\$1,751	741.5	9	69.7 LF

DESIGN READS

Frame	7 zones, 51'-4" along the house, 26'-6" at the deepest
Deck	74.0 LF of outer edge · 741.5 SF · 12'-0" above grade
Zone	L Stair pocket — 4'-4" x 4'-4" past the house's west wall; the stair leaves its north edge along the wall: 4'-4" x 4'-4", wall set back 4'-4"
Zone	P West pocket — 2'-4" beside the bay, ledgered to the recessed wall: 2'-4" x 10'-4", wall set back 4'-4"
Zone	P2 West strip — 4'-3" from the recessed wall to the curved front: 4'-3" x 16'-10", wall set back 4'-4"
Zone	W In front of the bay window — 7'-4" x 5', the bay projects 6' (multi-angle facets in the model): 7'-4" x 6'-6"
Zone	M Main run — 19'-11" wall, 11' deep: 19'-11" x 12'-6"
Zone	S Step — 2'-1" at the 7'-6" jog: 2'-1" x 18'-6", wall set back 7'-6"
Zone	A Alcove bay — 11'-1" into the 17' alcove, 26'-6" deep, ends at the east wing wall: 11'-1" x 26'-6", wall set back 15'-6"
Joists	57 2x12 #1 True Frame Joist SYP GC @ 12" OC
Rims	2-ply 2x12 front and sides and rear (freestanding)
Beam	DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb)
Beam	DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb)
Beam	DROP 6x12 DF A: 11'-0", 2 posts at 41'-3 1/2" / 50'-3 1/2" (worst 3,320 lb)
Posts	9 x 8x8 on 20" caissons 3'-2" deep

Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Vintage Coastline — 55 rows parallel to the house, picture frame & dividers in Vintage Dark Hickory, 3/16" gaps, Cortex
Rail	Fulton 36" Black: 13 bays, 17 posts, 69.7 LF
Stair	wall:L: 13 risers @ 7.38", 11 treads, 5 stringers, 4'-0" wide — 7 risers + 6 risers (left); landings 4'-0" x 4'-0" + 7'-8" x 5'-8"; landing guard 21 LF

- Rows per panel: panel 1 (0"-18'-3") 34 rows · panel 2 (18'-3"-38'-2") 26 rows · panel 3 (38'-2"-51'-4") 55 rows · 3 ripped pieces · 2 breakers
- 3 slivers under 6" beside the breakers / bay are not laid as boards (the border or breaker covers)
- DROP 6x12 DF L+P: 6'-7" long, 2 posts at 1'-0 1/2" / 5'-7 1/2"
- DROP 6x12 DF P2+W+M+S+A: 44'-7" long, 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; pieces 11'-9 1/2" · 20'-0" · 12'-9 1/2"
- DROP 6x12 DF A: 11'-0" long, 2 posts at 41'-3 1/2" / 50'-3 1/2"
- L: freestanding zone framed on its declared beam; no ledger — lateral bracing per the engineer
- P: 6'-8 3/4" cantilever past the drop beam exceeds L/4 of the 3'-5 3/4" back-span — engineered condition (owner-directed)
- P2: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 12'-11 3/4" back-span — engineered condition (owner-directed)
- W: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 2'-7 3/4" back-span — engineered condition (owner-directed)
- M: 3'-8 3/4" cantilever past the drop beam exceeds L/4 of the 8'-7 3/4" back-span — engineered condition (owner-directed)
- A: added a drop beam at 12'-2 3/8" from the house — 2x12 @ 12" OC SYP only spans 18'-0"

SITE & CODE FLAGS internal

ENGINEER	ENGINEERING	DROP 6x12 DF: 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm; 6x12 not in IRC R507.5 — treated as (3)2x12; engineer to confirm IRC Table R507.5
ENGINEER	ENGINEERING	Cantilever 8 3/4" > L/4 of the 0" back-span — owner-directed, in the stamped set IRC R507.5 / R507.6
ENGINEER	ENGINEERING	deck surface 12'-0" above grade — over 10': lateral bracing and an engineered design
CODE	ENGINEERING	posts over 8' (9'-5 1/2") — add diagonal knee bracing at each post (both directions at the corners) DCA6 / IRC R507.4
CODE	ENGINEERING	freestanding deck — no ledger lateral tie; provide diagonal bracing at posts or an engineered lateral system IRC R507.9 / R507.4
CODE	FIRE	WUI zone: Vintage is Ignition Resistant, Class A, WUI — keep the under-deck area clear of combustibles, 1-hr or noncombustible enclosure may be required IWUIC 504.7

CODE	STAIRS	wall:L stair: 13 risers — graspable handrail 34–38" on at least one side (IRC R311.7.8) IRC R311.7
CODE	STAIRS	wall:L stair landing: 36" min in the direction of travel and at least as wide as the stair; guard on every open side over 30" (21 LF counted) IRC R311.7.6 / R312.1
CHECK	SETBACK	rear setback not entered — confirm the required rear setback and the deck-to-line distance before the frame is laid out zoning
CHECK	SETBACK	side setback not entered — confirm the required side setback and the deck-to-line distance before the frame is laid out zoning
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	FIRE	Fulton rail is noncombustible (Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall)
INFO	STAIRS	wall:L stair: flights: 7 risers / 6 treads, run 5.8' · 6 risers / 5 treads, run 4.9' — left at the landing; landing guard 21 LF IRC R311.7
INFO	STAIRS	wall:L stair: 13 risers @ 7.38" · 11 treads @ 11.66" · run 10'-8 1/4" · stringers 5 @ 12" OC (composite treads) — 7 risers (run 5'-9 15/16") then 6 risers (run 4'-10 5/16") with a landing between IRC R311.7.5 / TimberTech

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
12	13	ea	2x12x10 #1 True Frame Joist SYP GC	+1 cull (field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 5 stair stringer)
30	31	ea	2x12x12 #1 True Frame Joist SYP GC	+1 cull (4 front rim ply · ledger · ledger, rear rim ply · rear rim ply, stair stringer · 2 front rim ply, stair stringer · ledger, stair stringer · blocking, field joist, stair stringer · 2 blocking, field joist · 2 PF joist, blocking, field joist · 2 blocking, field joist, left rim ply · 2 PF joist, blocking, right rim ply · blocking, front rim ply, left rim ply · blocking, front rim ply, ledger, left rim ply · blocking, front rim ply, ledger, right rim ply · blocking, right rim ply · 7 blocking)
26	27	ea	2x12x14 #1 True Frame Joist SYP GC	+1 cull (19 field joist · 2 PF joist · divider doubler · 2 left rim ply · 2 right rim ply)
20	21	ea	2x12x20 #1 True Frame Joist SYP GC	+1 cull (4 field joist · 4 PF joist · 4 left rim ply · 4 right rim ply · 2 front rim ply · ledger · divider doubler)
16	17	ea	2x12x24 #1 True Frame Joist SYP GC	+1 cull (10 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply)
2	3	ea	2x6x12 #1 True Frame Joist SYP GC	+1 cull (2 stair kicker/hanger board)
3	3	ea	6x12x12 Douglas Fir #2	exact (3 drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)

NET	ORDER	UNIT	ITEM	WHY
1	1	ea	6x12x20 Douglas Fir #2	exact (drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)
1	1	ea	6x12x16 Douglas Fir #2	exact (drop beam: DROP 6x12 DF L+P · DROP 6x12 DF P2+W+M+S+A · DROP 6x12 DF A)
9	9	ea	8x8x10 Douglas Fir #1 timber (S4S, end grain sealed)	exact (yields 9 posts at ~9'-5 1/2")

Footings

NET	ORDER	UNIT	ITEM	WHY
3	3	ea	Sonotube 20" x 12' (cut to length — 3 caissons per tube) — caissons 20" x 3'-2" + 6" above grade (frost 3'-0")	exact (9 caissons, 3 per tube)
126	126	bag	Quikrete 80# concrete mix	exact (14 bags per caisson)
7	7	ea	#4 rebar x 20' (4 verticals per caisson)	exact — engineer governs
8	8	ea	#3 rebar x 20' (ties)	exact — engineer governs
9	9	ea	5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)	exact
9	9	ea	Simpson ABU88Z standoff post base 8x8	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
114	114	ea	Simpson LUS210Z face-mount hanger 2x10/2x12	exact (57 joists x 2 ends)
28	28	ea	Simpson HUCQ212-2-SDS concealed-flange double 2x12 hanger	exact (2 side rims x 2 ends) — confirm stock
3	3	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	912 needed of ~1050
3	3	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	1534 needed of ~1950
1	1	box	FastenMaster LedgerLOK 3-5/8" 50 ct	48 needed
28	28	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
85	85	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (57 joists + 2 rims at each drop beam)
9	9	ea	Simpson CCQ68SDS2.5 column cap 6x beam / 8x post	exact
1	1	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	90 needed (bases 0 · caps 90)
10	10	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	998 needed (laminations 554 · blocking 444)
46	48	ea	7/16" x 4-1/2" HDG hex bolt + nut + 2 washers	+2 (23 rail posts x 2)
10	10	ea	Simpson LSCZ adjustable stair-stringer connector	exact (wall:L stair, 1 per stringer at the rim — 2 flights)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
2	2	roll	Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)	79 LF of house wall with laps — nothing on a privacy wall
8	8	ea	Aluminum L-flashing 10' (over ledger + side rims)	house walls only
8	8	ea	26 ga galvanized wall flashing 12" x 10' (CWRC 404.3.2: noncombustible flashing at the deck/wall intersection, behind cladding)	WUI practice — house walls only
16	16	roll	Nichigo G-Tape 3035 2" x 65' black	977' needed of 1040'
5	5	roll	Nichigo G-Tape 3035 4" x 65' black	303' needed of 325'

Decking

NET	ORDER	UNIT	ITEM	WHY
30	32	ea	TimberTech Vintage Coastline 1x6x12 Square Edge	+2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)
30	32	ea	TimberTech Vintage Coastline 1x6x16 Square Edge	+2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)
40	42	ea	TimberTech Vintage Coastline 1x6x20 Square Edge	+2 (30 x 12' · 30 x 16' · 40 x 20' (17 cut into 2-3 rows) · panel 1 34 rows · panel 2 26 rows · panel 3 55 rows · every row one piece)
1	1	ea	TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers	exact (border 24'-8",12'-4 9/16" 2'-8 1/2" + border 4'-4",6'-0" 2'-4 15/16" + border 20'-2",12'-5 5/8" 2'-4" + border 22'-5",12'-5 5/8" 2'-4")
5	6	ea	TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers	+1 (breaker DM/S 19'-11 5/8" · border 38'-2",11'-0" 13'-2 15/16" + border 4'-4",0" 6'-1" · breaker DW/M 12'-7" + border 6'-8",6'-0" 5'-1" · border 0",-4'-4" 4'-4 15/16" + border 0",0" 4'-4 15/16" + border 35'-11",11'-0" 2'-8 15/16" + border 6'-8",11'-0" 2'-8 15/16" + border 8'-11",11'-4 13/16" 2'-8 13/16" + border 33'-8",11'-4 13/16" 2'-8 13/16" · border 11'-2",11'-8 7/8" 2'-8 11/16" + border 31'-5",11'-8 7/8" 2'-8 11/16" + border 13'-5",12'-0 1/8" 2'-8 5/8" + border 29'-2",12'-0 1/8" 2'-8 5/8" + border 15'-8",12'-2 3/4" 2'-8 9/16" + border 26'-11",12'-2 3/4" 2'-8 9/16" + border 17'-11",12'-4 9/16" 2'-8 1/2")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
10	10	box	Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)	3454 screws (2 per board at every bearing point) · 3500 supplied
1	1	box	Cortex for TimberTech PVC, Dark Hickory plugs — borders + dividers	232 screws

Fascia

NET	ORDER	UNIT	ITEM	WHY
21	22	ea	TimberTech Vintage Fascia Coastline 0.5x11.75x12	+1 (left fascia (L) 4'-4 1/2" · right fascia (L) 4'-4 1/2" · front fascia (L) 4'-5" · left fascia (P) 10'-4 1/2" · right fascia (P) 10'-4 1/2" · front fascia (P) 2'-5" · left fascia (P2) 12'-0" · left fascia (P2) 4'-10 1/2" · right fascia (P2) 12'-0" · right fascia (P2) 4'-10 1/2" · front fascia (P2) 4'-4" · left fascia (W) 6'-6 1/2" · right fascia (W) 6'-6 1/2" · front fascia (W) 7'-5" · left fascia (M) 12'-0" · left fascia (M) 6 1/2" · right fascia (M) 12'-0" · right fascia (M) 6 1/2" · front fascia (M) 12'-0" · front fascia (M) 8'-0" · left fascia (S) 12'-0" · left fascia (S) 6'-6 1/2" · right fascia (S) 12'-0" · right fascia (S) 6'-6 1/2" · front fascia (S) 2'-2" · left fascia (A) 12'-0" · left fascia (A) 12'-0" · left fascia (A) 2'-6 1/2" · right fascia (A) 12'-0" · right fascia (A) 12'-0" · right fascia (A) 2'-6 1/2" · front fascia (A) 11'-2")

Rail

NET	ORDER	UNIT	ITEM	WHY
7	7	ea	TimberTech Fulton Rail 6' x 36" level panel Black	exact (cut to 1'-8 1/2" / 10 1/2" / 3'-8 1/2" / 5'-4 1/2" / 5'-5 3/4")
6	6	ea	TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)	exact (cut to 5'-10 1/2" / 6'-2" / 6'-4 5/16")
4	4	ea	TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt	exact
8	8	ea	TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt	exact
5	5	ea	TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt	exact
1	1	ea	Rust-Oleum flat black touch-up (cut ends)	exact

Stairs

NET	ORDER	UNIT	ITEM	WHY
5	5	ea	2x12x12 #1 True Frame Joist SYP GC — landing 1 frame (4'-0" x 4'-0", wall:L stair)	+1 (rims + 6 joists @ 12")
4	4	ea	8x8x8 — landing 1 posts (4)	exact
4	4	ea	Landing 1 footings — 4 concrete piers 12" x frost, wet-set bases	exact
4	4	ea	TimberTech Vintage Coastline 1x6x12 — landing 1 decking (16 SF)	+10% cuts
8	8	ea	2x12x12 #1 True Frame Joist SYP GC — landing 2 frame (7'-8" x 5'-8", wall:L stair)	+1 (rims + 9 joists @ 12")
4	4	ea	8x8x8 — landing 2 posts (4)	exact
4	4	ea	Landing 2 footings — 4 concrete piers 12" x frost, wet-set bases	exact
9	9	ea	TimberTech Vintage Coastline 1x6x12 — landing 2 decking (43 SF)	+10% cuts
11	12	ea	TimberTech Vintage Coastline 1x6x12 Square Edge — treads (wall:L stair)	+1 (11 treads x 2 boards @ 4'-0", 2 per board)
7	7	ea	TimberTech Vintage Coastline riser board 7-1/4" x 12' (wall:L stair)	exact (13 risers @ 4'-0")
2	2	ea	TimberTech Fulton Rail 6' x 36" STAIR panel Black (wall:L stair)	exact

NET	ORDER	UNIT	ITEM	WHY
1	1	ea	TimberTech Fulton Rail 8' x 36" STAIR panel Black (wall:L stair)	exact
6	6	ea	TimberTech Fulton 2" STAIR post Black w/ brackets, cap (wall:L stair)	exact (top + bottom of each rail side, 2 flights)
3	3	ea	TimberTech Fulton Rail 8' x 36" LEVEL panel Black — landing guard (wall:L stair, 21 LF)	exact (open sides of the landings)
5	5	ea	TimberTech Fulton 2" post Black w/ brackets, cap — landing guard (wall:L stair)	exact (bay ends + turn corners)
1	1	ea	Fulton graspable handrail kit (wall:L stair)	exact (4+ risers — IRC R311.7.8)

Site

NET	ORDER	UNIT	ITEM	WHY
700	700	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Ledger	2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider)
Rail	Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists (L)	2x12	3'-10"	3	@ 12" OC, crown up
Picture-frame joists (L)	2x12	3'-10"	2	centre 5 1/4" from each frame face
Side rim plies (L)	2x12	3'-10"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (L)	2x12	4'-4"	2	inner ply takes the hangers; outer ply laminated after
Rear rim plies (freestanding) (L)	2x12	4'-4"	2	
Blocking (L)	2x12	10 1/2"	8	2 rows at 3'-7 1/4", 3'-7 1/4" from the wall
Field joists (P)	2x12	9'-11 1/2"	1	@ 12" OC, crown up
Picture-frame joists (P)	2x12	9'-11 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (P)	2x12	9'-11 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered

MEMBER	STOCK	LENGTH	QTY	NOTE
Front rim plies (P)	2x12	2'-4"	2	inner ply takes the hangers; outer ply laminated after
Ledger (P)	2x12	2'-4"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (P)	2x12	10 1/2"	4	2 rows at 3'-7 1/4", 1'-10 3/8" from the wall
Field joists (P2)	2x12	16'-5 1/2"	3	@ 12" OC, crown up
Picture-frame joists (P2)	2x12	16'-5 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (P2)	2x12	16'-5 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (P2)	2x12	4'-3"	2	inner ply takes the hangers; outer ply laminated after
Ledger (P2)	2x12	4'-3"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (P2)	2x12	10 1/2"	8	2 rows at 13'-1 1/4", 6'-7 3/8" from the wall
Field joists (W)	2x12	6'-1 1/2"	6	@ 12" OC, crown up
Picture-frame joists (W)	2x12	6'-1 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (W)	2x12	6'-1 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (W)	2x12	7'-4"	2	inner ply takes the hangers; outer ply laminated after
Ledger (W)	2x12	7'-4"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (W)	2x12	10 1/2"	14	2 rows at 2'-9 1/4", 1'-5 3/8" from the wall
Field joists (M)	2x12	12'-1 1/2"	19	@ 12" OC, crown up
Picture-frame joists (M)	2x12	12'-1 1/2"	2	centre 5 1/4" from each frame face
Divider doublers (M)	2x12	12'-1 1/2"	1	sister a joist on the divider line so both board edges bear
Side rim plies (M)	2x12	12'-1 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (M)	2x12	19'-11"	2	inner ply takes the hangers; outer ply laminated after
Ledger (M)	2x12	19'-11"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (M)	2x12	10 1/2"	40	2 rows at 8'-9 1/4", 4'-5 3/8" from the wall
Field joists (S)	2x12	18'-1 1/2"	1	@ 12" OC, crown up
Picture-frame joists (S)	2x12	18'-1 1/2"	2	centre 5 1/4" from each frame face
Divider doublers (S)	2x12	18'-1 1/2"	1	sister a joist on the divider line so both board edges bear
Side rim plies (S)	2x12	18'-1 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (S)	2x12	2'-1"	2	inner ply takes the hangers; outer ply laminated after

MEMBER	STOCK	LENGTH	QTY	NOTE
Ledger (S)	2x12	2'-1"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (S)	2x12	10 1/2"	4	2 rows at 16'-3 1/4", 8'-2 3/8" from the wall
Field joists (A)	2x12	26'-1 1/2"	10	@ 12" OC, crown up
Picture-frame joists (A)	2x12	26'-1 1/2"	2	centre 5 1/4" from each frame face
Side rim plies (A)	2x12	26'-1 1/2"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies (A)	2x12	11'-1"	2	inner ply takes the hangers; outer ply laminated after
Ledger (A)	2x12	11'-1"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (A)	2x12	10 1/2"	33	3 rows at 12'-2 3/8", 24'-3 1/4", 6'-1 15/16" from the wall
Field boards panel 1	Vintage Coastline	7"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	9"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	11"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	1'-0"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	1'-7"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	2'-3"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	2'-11"	1	one board wall to border, no joints
Field boards panel 1 (rip)	Vintage Coastline	3'-0"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 1	Vintage Coastline	6'-2"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	6'-4"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	6'-5"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	6'-7"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	6'-9"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	6'-11"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	7'-0"	2	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	7'-2"	2	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	7'-4"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	7'-5"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	8'-0"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	8'-8"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	10'-1"	1	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	10'-6"	8	one board wall to border, no joints
Field boards panel 1	Vintage Coastline	10'-11"	10	one board wall to border, no joints

MEMBER	STOCK	LENGTH	QTY	NOTE
Field boards panel 2 (rip)	Vintage Coastline	10'-0"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 2	Vintage Coastline	14'-8"	1	one board wall to border, no joints
Field boards panel 2	Vintage Coastline	17'-10"	1	one board wall to border, no joints
Field boards panel 2	Vintage Coastline	19'-5"	22	one board wall to border, no joints
Field boards panel 2 (rip)	Vintage Coastline	19'-5"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 3	Vintage Coastline	11'-1"	17	one board wall to border, no joints
Field boards panel 3	Vintage Coastline	12'-11"	38	one board wall to border, no joints
Slivers under 6" (not laid)	Vintage Coastline	0"	3	beside the breakers at the bay — the breaker / border covers
Stair stringers (wall:L, flight 1)	2x12	8'-3"	5	7 risers @ 7.385" · 6 treads @ 11.66" · stringers 12" OC
Stair stringers (wall:L, flight 2)	2x12	7'-1 3/16"	5	6 risers @ 7.385" · 5 treads @ 11.66" · stringers 12" OC
DROP 6x12 DF L+P beam	6x12	6'-7"	1	y -8 13/16"; posts at 1'-0 1/2" / 5'-7 1/2"
Posts under DROP 6x12 DF L+P	8x8	9'-5 1/2"	2	field-measure each
DROP 6x12 DF P2+W+M+S+A beam	6x12	44'-7"	3	y 8'-9 3/16"; posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; pieces 11'-9 1/2" · 20'-0" · 12'-9 1/2", spliced over posts
Posts under DROP 6x12 DF P2+W+M+S+A	8x8	9'-5 1/2"	5	field-measure each
DROP 6x12 DF A beam	6x12	11'-0"	1	y -3'-3 5/8"; posts at 41'-3 1/2" / 50'-3 1/2"
Posts under DROP 6x12 DF A	8x8	9'-5 1/2"	2	field-measure each
Border 0",-4'-4"	Vintage Dark Hickory square	4'-4 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 0",0"	Vintage Dark Hickory square	4'-4 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 4'-4",0"	Vintage Dark Hickory square	6'-1"	1	mitre at the corners; dividers on the doubled joist
Border 4'-4",6'-0"	Vintage Dark Hickory square	2'-4 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 6'-8",6'-0"	Vintage Dark Hickory square	5'-1"	1	mitre at the corners; dividers on the doubled joist
Border 6'-8",11'-0"	Vintage Dark Hickory square	2'-8 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 8'-11",11'-4 13/16"	Vintage Dark Hickory square	2'-8 13/16"	1	mitre at the corners; dividers on the doubled joist

MEMBER	STOCK	LENGTH	QTY	NOTE
Border 11'-2",11'-8 7/8"	Vintage Dark Hickory square	2'-8 11/16"	1	mitre at the corners; dividers on the doubled joist
Border 13'-5",12'-0 1/8"	Vintage Dark Hickory square	2'-8 5/8"	1	mitre at the corners; dividers on the doubled joist
Border 15'-8",12'-2 3/4"	Vintage Dark Hickory square	2'-8 9/16"	1	mitre at the corners; dividers on the doubled joist
Border 17'-11",12'-4 9/16"	Vintage Dark Hickory square	2'-8 1/2"	1	mitre at the corners; dividers on the doubled joist
Border 20'-2",12'-5 5/8"	Vintage Dark Hickory square	2'-4"	1	mitre at the corners; dividers on the doubled joist
Border 22'-5",12'-5 5/8"	Vintage Dark Hickory square	2'-4"	1	mitre at the corners; dividers on the doubled joist
Border 24'-8",12'-4 9/16"	Vintage Dark Hickory square	2'-8 1/2"	1	mitre at the corners; dividers on the doubled joist
Border 26'-11",12'-2 3/4"	Vintage Dark Hickory square	2'-8 9/16"	1	mitre at the corners; dividers on the doubled joist
Border 29'-2",12'-0 1/8"	Vintage Dark Hickory square	2'-8 5/8"	1	mitre at the corners; dividers on the doubled joist
Border 31'-5",11'-8 7/8"	Vintage Dark Hickory square	2'-8 11/16"	1	mitre at the corners; dividers on the doubled joist
Border 33'-8",11'-4 13/16"	Vintage Dark Hickory square	2'-8 13/16"	1	mitre at the corners; dividers on the doubled joist
Border 35'-11",11'-0"	Vintage Dark Hickory square	2'-8 15/16"	1	mitre at the corners; dividers on the doubled joist
Border 38'-2",11'-0"	Vintage Dark Hickory square	13'-2 15/16"	1	mitre at the corners; dividers on the doubled joist
Breaker dw/m	Vintage Dark Hickory square	12'-7"	1	mitre at the corners; dividers on the doubled joist
Breaker dm/s	Vintage Dark Hickory square	19'-11 5/8"	1	mitre at the corners; dividers on the doubled joist
Stair treads (wall:L)	Vintage Coastline square	4'-0"	22	2 boards per tread, nosing 3/4" over the riser
Stair risers (wall:L)	riser board	4'-0"	13	rip to 7.38" less tread thickness
Left fascia (l)	fascia	4'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (l)	fascia	4'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (l)	fascia	4'-5"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (p)	fascia	10'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house

MEMBER	STOCK	LENGTH	QTY	NOTE
Right fascia (p)	fascia	10'-4 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (p)	fascia	2'-5"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (p2)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (p2)	fascia	4'-10 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (p2)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (p2)	fascia	4'-10 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (p2)	fascia	4'-4"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (w)	fascia	6'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (w)	fascia	6'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (w)	fascia	7'-5"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (m)	fascia	6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (m)	fascia	6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	8'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (s)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (s)	fascia	6'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (s)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (s)	fascia	6'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (s)	fascia	2'-2"	1	flush under the board nose; full board at the front corners, short piece at the house

MEMBER	STOCK	LENGTH	QTY	NOTE
Left fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (a)	fascia	2'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (a)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (a)	fascia	2'-6 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (a)	fascia	11'-2"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK internal

Materials (raw)	\$42,995.79
Materials × 1.15	\$49,445.16
Sales tax 7.90% on factored materials	\$3,906.17
Labor (rate card)	\$25,056.20
Work — carries the margin	\$78,407.53
General conditions at cost	\$2,966.00
Cost	\$81,373.53
Sell — check or ACH at 42.9% GM	\$140,327 · \$189.25/SF
Sell at the 40% floor	\$133,645
Financed	\$150,889 · \$1,751/mo at 6.99% × 10 yr
Gross profit	\$58,953
Engineering — outside the price, at cost	\$1,800–\$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	700	SF	\$2.00	\$1,400.00
Caissons 20" x 3.2' (frost 36")	9	hole	\$150.00	\$1,350.00
Large footings 20" add-on	9	footing	\$150.00	\$1,350.00
Frame	741.5	SF	\$10.00	\$7,415.00
Decking & fascia	741.5	SF	\$8.00	\$5,932.00

LINE	QTY	UNIT	RATE	EXT
Railing	69.7	LF	\$20.00	\$1,394.00
Extra beam set	2	set	\$200.00	\$400.00
Stairs (wall:L) — 2 flights	13	riser	\$150.00	\$1,950.00
Landing guard rail (wall:L stair)	21	LF	\$20.00	\$420.00
Stair landing 4'-0" x 4'-0" — frame, deck, 4 footings	16	SF	\$58.00	\$928.00
Stair landing 7'-8" x 5'-8" — frame, deck, 4 footings	43.4	SF	\$58.00	\$2,517.20

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	\$2,966	\$4/SF x 741.5 SF

What the price includes — sell values

ITEM	VALUE
Demolition & haul-off	\$2,577
Foundation & structure	\$56,207
Decking & fascia	\$48,135
Railing	\$9,354
Stairs	\$20,938
Site & project services	\$3,116
Total, check or ACH	\$140,327
Total, financed	\$150,889

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

GLENGARRY PL UPPER DECK — 2: FULTON RAIL · NEW CAISSONS AND 8X8 DF POSTS

2x12 SYP frame at 12", 6x12 beam, 8x8 posts — on 9 caissons. TimberTech Vintage Coastline decking with a Dark Hickory border, Fulton rail.

741.5 SF · 7 zones · **12'-0"** above grade, engineered · **30 psf** snow design

Prepared for the owner of 166 Glengarry Pl, Castle Rock

PRICE

Total investment \$150,889 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$1,751/month

Check or ACH: \$140,327 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	\$2,577
Foundation & structure	\$56,207
Decking & fascia	\$48,135
Railing	\$9,354
Stairs	\$20,938
Site & project services	\$3,116
Total, check or ACH	\$140,327
Total, financed	\$150,889

Engineering — \$1,800 to \$3,000. Stamped drawings by a Colorado engineer (30 psf snow, caissons). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT SELECTION

Decking	TimberTech PVC Vintage, Coastline; picture frame and dividers in Vintage Dark Hickory. Square-shoulder boards, 3/16" gaps, matching fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Railing	TimberTech Fulton Rail, Black, 36". 69.7 LF.
Frame	57 2x12 #1 True Frame Joist SYP GC @ 12" OC; 2-ply 2x12 front and sides and rear (freestanding); DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb); DROP 6x12 DF A: 11'-0", 2 posts at 41'-3 1/2" / 50'-3 1/2" (worst 3,320 lb).

ELEMENT SELECTION

Foundation 20" caissons 3'-2" deep (frost depth 3'-0").

Design load 30 psf snow, stamped by a Colorado engineer. Colorado Wildfire Resiliency Code practice: Class A decking, noncombustible rail, metal flashing at every wall.

Hardware Simpson ZMAX galvanized connectors. Cortex fasteners with color-matched plugs.

PLAN

7 zones · 741.5 SF · 51'-4" along the house · 12'-0" above grade · 69.7 LF Black rail

- **L · Stair pocket — 4'4" x 4'4" past the house's west wall; the stair leaves its north edge along the wall — 4'-4" x 4'-4".**
- **P · West pocket — 2'4" beside the bay, ledgered to the recessed wall — 2'-4" x 10'-4".**
- **P2 · West strip — 4'3" from the recessed wall to the curved front — 4'-3" x 16'-10".**
- **W · In front of the bay window — 7'4" x 5', the bay projects 6' (multi-angle facets in the model) — 7'-4" x 6'-6".**
- **M · Main run — 19'11" wall, 11' deep — 19'-11" x 12'-6".**
- **S · Step — 2'1" at the 7'6" jog — 2'-1" x 18'-6".**
- **A · Alcove bay — 11'1" into the 17' alcove, 26'6" deep, ends at the east wing wall — 11'-1" x 26'-6", no rail on the wall side.**

Zones are named left to right standing in the yard facing the house.

- Stair wall:L: 13 risers @ 7.38", 11 treads, 5 stringers, 4'-0" wide — 7 risers + 6 risers (left); landings 4'-0" x 4'-0" + 7'-8" x 5'-8"; landing guard 21 LF

SCHEDULE

1. Engineering & ordering (2-3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Days 2-3 — layout, 9 caissons augered and poured.
4. Days 4-5 — concrete cure.
5. Days 6-10 — posts and beams set.
6. Days 11-16 — frame, flashing at every wall.
7. Days 17-20 — decking, borders first.
8. Days 21-22 — Fulton rail.
9. Day 23 — stairs.
10. Day 24 — final clean; end of construction.

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer

and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

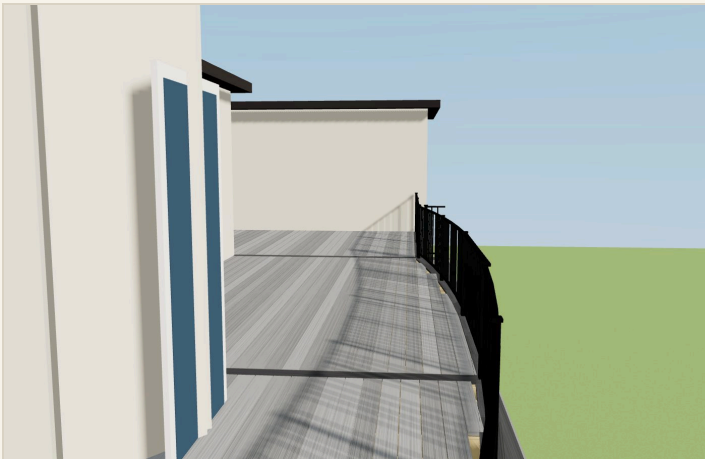
RENDERS 3D model • design intent • timbers shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings

ISOMETRICS



Isometric from the stair corner



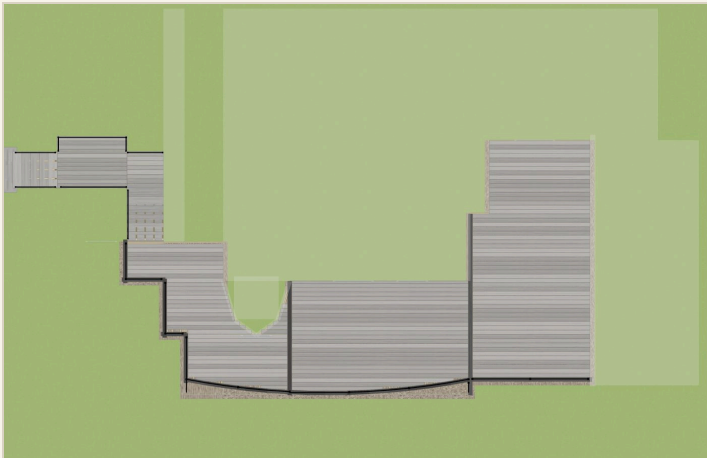
Isometric from the open corner



Low isometric

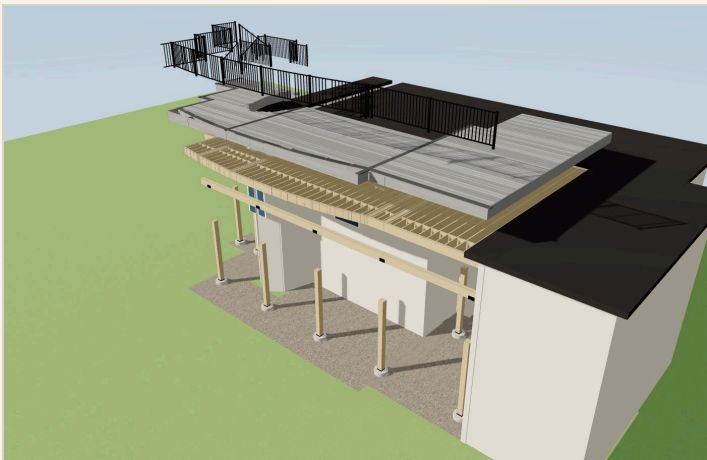


Isometric from under the frame



Plan from above

EXPLODED



Exploded — every layer



Exploded — from the open corner



Exploded — low angle

DRAWING SET G · S · A · D sheets — permit set with G-001 on top

G-001

General notes · design criteria

G-001

GENERAL NOTES · DESIGN CRITERIA

Glengarry PI upper deck — 2: Fulton rail · new caissons and 8x8 DF...

GSX · Jade Pender 303-550-9558 · September 21, 2026 · Rev 0 FOR REVIEW

PROJECT

Glengarry PI upper deck — 2: Fulton rail · new caissons and 8x8 DF posts
166 Glengarry Pl, Castle Rock CO 80108
741.5 SF · 12'-0" above grade · 7 zones

DESIGN CRITERIA

Deck live load

Ground snow

Dead load

Roof snow (cover)

Design total

Wind

Frost depth

Soil bearing

Seismic

Wildfire (WUI)

Codes

Engineering

STRUCTURE

Footings

Posts

Beams

Joists

Rims

Ledger

Decking

Rail

SHEET INDEX

G-001

A-001

S-100

S-101

A-101

A-201

A-301

D-501

T-400

T-700

SITE

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER

ENGINEER

ENGINEER

CODE

CODE

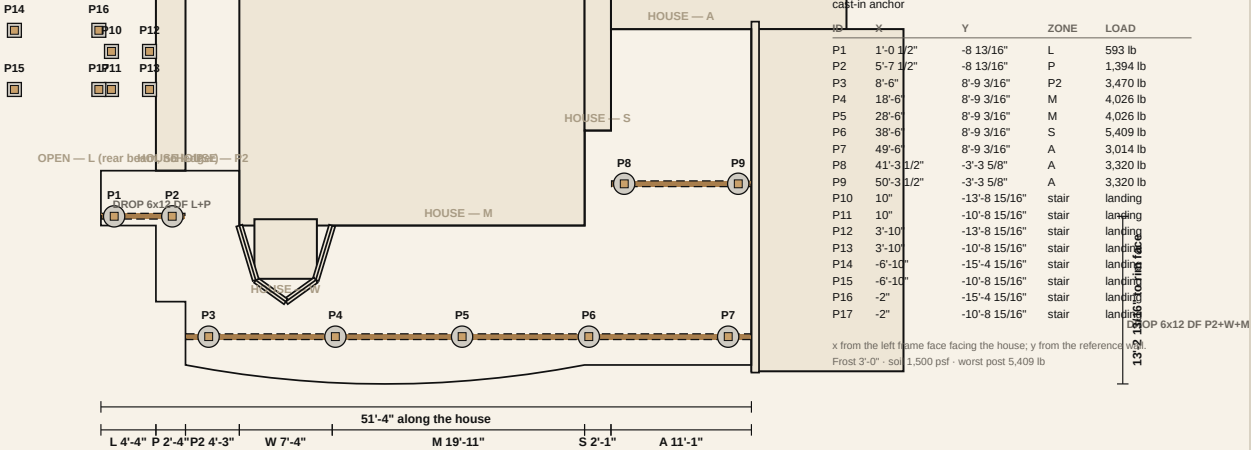
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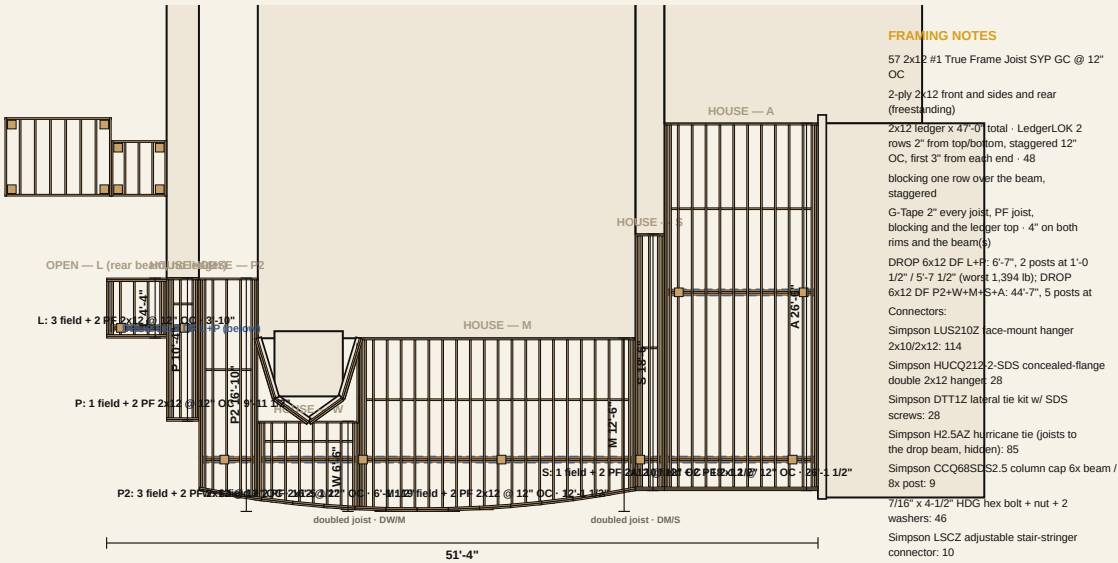
CODE

CODE

No scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.





A-101 Decking plan

A-101

DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

Glengarry Pl upper deck — 2: Fulton rail · new caissons and 8x8 DF...

GSX · Jade Pender 303-550-9558 · September 21, 2026 · Rev 0 FOR REVIEW



Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

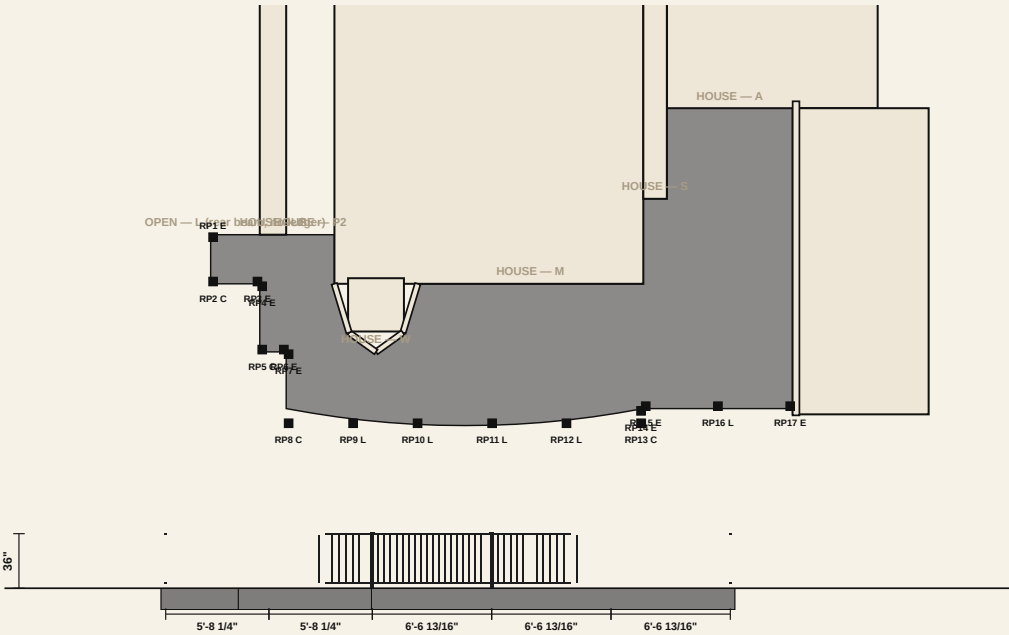
RAILING

Fulton 36" Black: 13 bays, 17 posts,
69.7 LF

Fulton 36" — 17 posts (4 CORNER, 8 END,
5 LINE) inside the outer rim ply, inner
ply pocketed 2" wide, (2) 7/16" x 4-1/2"
bolts per post

Bays: end:left 3'-11" · front:L 3'-11" ·
step:L-P 5'-7" · front:P 1'-11"
step:P-P2 6'-1" · front:P2+W+M 5'-8 1/4"
· front:P2+W+M 5'-8 1/4" · front:P2+W+M
6'-6 13/16" · front:P2+W+M 6'-6 13/16" ·
front:P2+W+M 6'-6 13/16" · step:M-S

Guard 36" min above the deck; openings
under 4" (IRC R312). Rail LF for labor =
deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

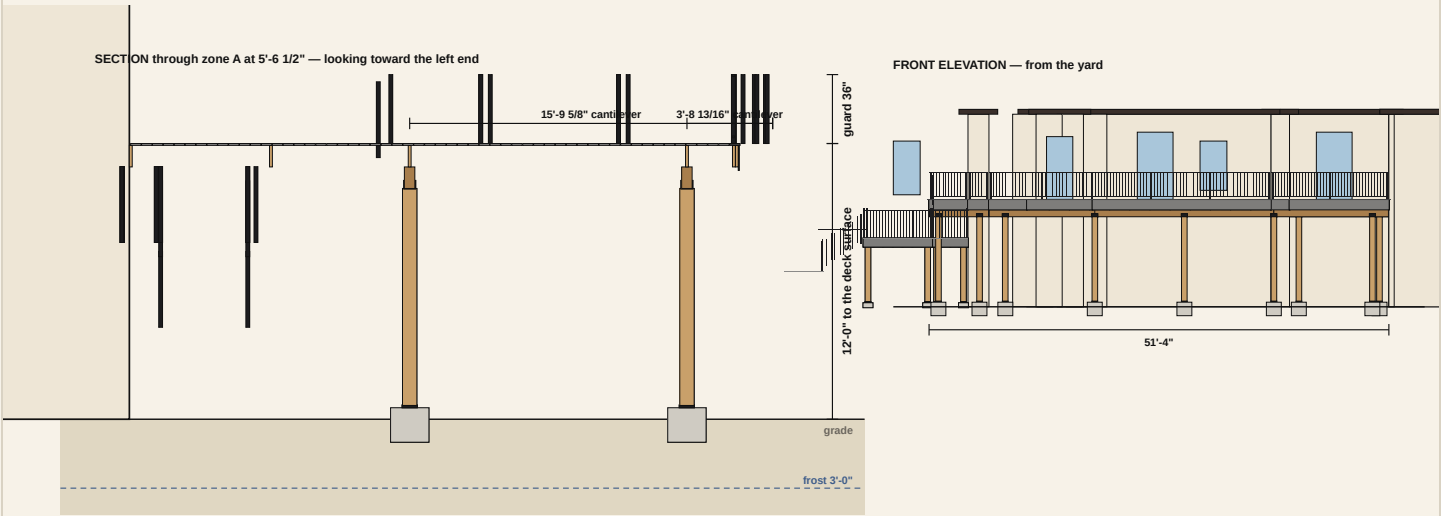
A-301

SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

Glengarry Pl upper deck — 2: Fulton rail · new caissons and 8x8 DF...

GSX · Jade Pender 303-550-9558 · September 21, 2026 · Rev 0 FOR REVIEW



SECTION NOTES

Deck surface 12'-0" above grade; guard 36" where the drop exceeds 30"

DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0 1/2" / 5'-7 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6" (worst 5,409 lb);

Posts 9 x 8x8 on 20" caissons 3'-2" deep

2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

1 LEDGER AT THE HOUSE

membrane 12" up the wall

decking

flashing over the ledger, under the siding

ledger

2x12 joist in LUS28Z

house rim

2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners · 26 ga wall flashing (WUJ)

2 POST · BEAM · CAP

joists bear on the beam, H2.5AZ each

6x12 drop beam

CCQ68SDS2.5

8x8 post

DROP 6x12 DF L+P: 6'-7", 2 posts at 1'-0" / 2' / 5'-7" 1/2" (worst 1,394 lb); DROP 6x12 DF P2+W+M+S+A: 44'-7", 5 posts at 8'-6" / 18'-6" / 28'-6" / 38'-

3 FOOTING · POST BASE

post

post base, 5/8" cast-in anchor grade

20" caisson

frost 3'-0"

4 RAIL POST · GUARD

top rail

Fulton post 36"

rim (2-ply, inner ply pocketed)

Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

5 DECK EDGE · BORDER

border

field · 3/16" gaps

rim

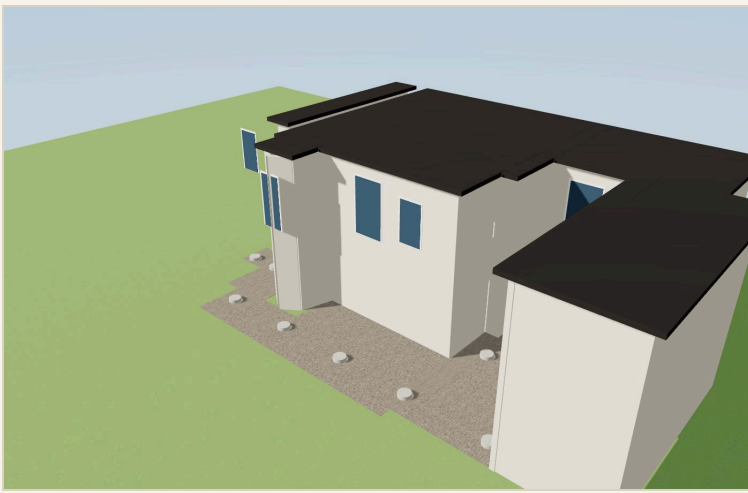
fascia

overhang 1"

Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.



Step 1 • Footings — 9 caissons 20" x 3.2' below frost

T-400 • Step 1 • Footings

GOES IN

- (3) Sonotube 20" x 12' (cut to length — 3 caissons per tube) — caissons 20" x 3'-2" + 6" above grade (frost 3'-0")
- (126) Quikrete 80# concrete mix
- (7) #4 rebar x 20' (4 verticals per caisson)
- (8) #3 rebar x 20' (ties)
- (9) 5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)
- (9) Simpson ABU88Z standoff post base 8x8

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall:
DROP 6x12 DF L+P at -8 13/16" from the reference wall, posts at 1'-0 1/2" / 5'-7 1/2";
DROP 6x12 DF P2+W+M+S+A at 8'-9 3/16" from the reference wall, posts at 8'-6" / 18'-6" / 28'-6" / 38'-6" / 49'-6"; DROP 6x12 DF A at -3'-3 5/8" from the reference wall, posts at 41'-3 1/2" / 50'-3 1/2".
- Auger 9 holes 20" x 3'-2" (frost 3'-0"), tube to 6" above grade, cage in, pour, set the 5/8" anchors wet. Cure 2 days.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Frost depth met
- Photos before backfill

T-401 · Step 2 · Posts

GOES IN

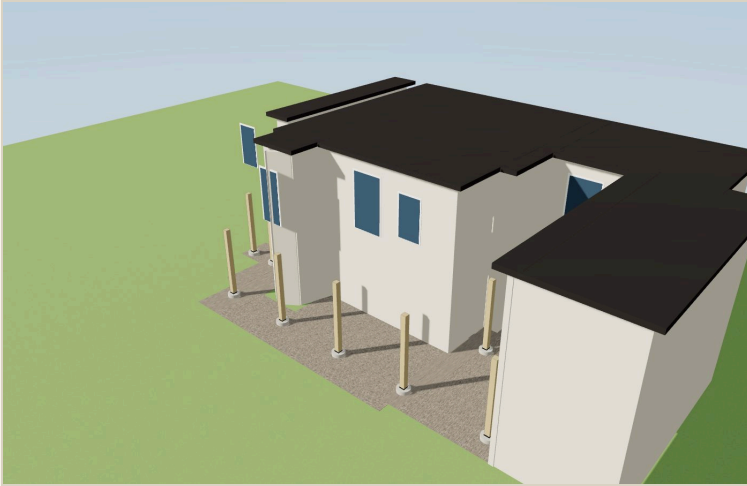
- (9) 8x8x10 Douglas Fir #1 timber (S4S, end grain sealed)
- (9) 5/8" x 8" HDG anchor bolt w/ nut + washer (cast-in)
- (9) Simpson ABU88Z standoff post base 8x8

HOW

- 8x8 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP 6x12 DF L+P posts 9'-5 1/2" / DROP 6x12 DF P2+W+M+S+A posts 9'-5 1/2" / DROP 6x12 DF A posts 9'-5 1/2" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 2 · Posts — 8x8 #2 GC on ZMAX bases

T-403 · Step 4 · Beams

GOES IN

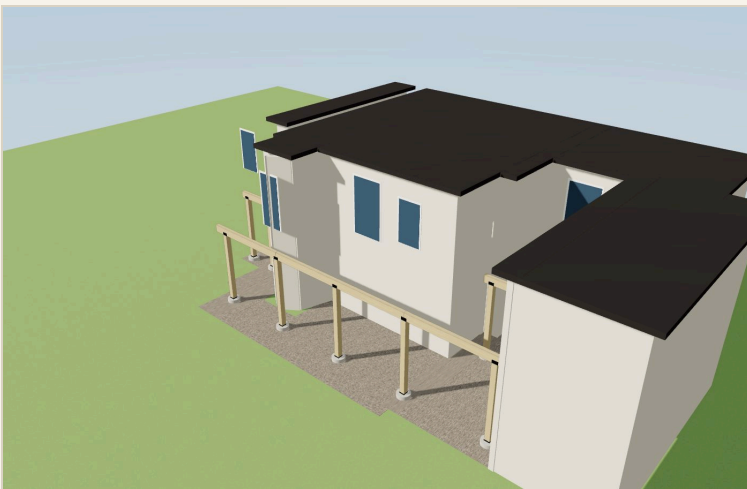
- (2) 2x6x12 #1 True Frame Joist SYP GC
- (3) 6x12x12 Douglas Fir #2
- (1) 6x12x20 Douglas Fir #2
- (1) 6x12x16 Douglas Fir #2
- (9) Simpson CCQ68SDS2.5 column cap 6x beam / 8x post

HOW

- DROP 6x12 DF L+P: 6'-7" one piece, 2 posts @ 4'-7" max
- DROP 6x12 DF P2+W+M+S+A: 44'-7" in 3 pieces spliced over posts, 5 posts @ 11'-0" max
- DROP 6x12 DF A: 11'-0" one piece, 2 posts @ 9'-0" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 4 · Beams — DROP 6x12 DF L+P (2 posts) · DROP 6x12 DF P2+W+M+S+A (5 posts) · DROP 6x12 DF A (2 posts)

T-404 · Step 5 · Frame

GOES IN

- (12) 2x12x10 #1 True Frame Joist SYP GC
- (30) 2x12x12 #1 True Frame Joist SYP GC
- (26) 2x12x14 #1 True Frame Joist SYP GC
- (20) 2x12x20 #1 True Frame Joist SYP GC
- (16) 2x12x24 #1 True Frame Joist SYP GC
- (3) 6x12x12 Douglas Fir #2
- (114) Simpson LUS210Z face-mount hanger 2x10/2x12
- (28) Simpson HUCQ212-2-SDS concealed-flange double 2x12 hanger
- (3) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (3) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (85) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (2) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (8) Aluminum L-flashing 10' (over ledger + side rims)

HOW

- 2x12 ledger x 47'-0" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 48 fasteners
- L: 3 joists @ 12" OC x 3'-10"; P: 1 joists @ 12" OC x 9'-11 1/2"; P2: 3 joists @ 12" OC x 16'-5 1/2"; W: 6 joists @ 12" OC x 6'-1 1/2"; M: 19 joists @ 12" OC x 12'-1 1/2"; S: 1 joists @ 12" OC x 18'-1 1/2"; A: 10 joists @ 12" OC x 26'-1 1/2". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 5 · Frame — 2x12 joists at 12", rims, ledger, blocking over the beam

T-405 · Step 6 · Decking

GOES IN

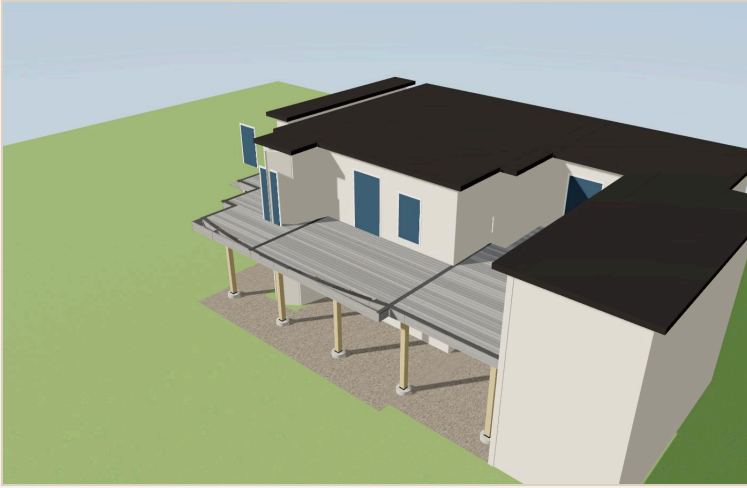
- (30) TimberTech Vintage Coastline 1x6x12 Square Edge
- (30) TimberTech Vintage Coastline 1x6x16 Square Edge
- (40) TimberTech Vintage Coastline 1x6x20 Square Edge
- (1) TimberTech Vintage Dark Hickory 1x6x16 Square Edge — borders & dividers
- (5) TimberTech Vintage Dark Hickory 1x6x20 Square Edge — borders & dividers
- (21) TimberTech Vintage Fascia Coastline 0.5x11.75x12
- (10) Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)

HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (3454 field + 232 border/divider)
- Borders first, then the field.
- Rows per panel: panel 1 (0"-18'-3") 34 rows · panel 2 (18'-3"-38'-2") 26 rows · panel 3 (38'-2"-51'-4") 55 rows · 3 ripped pieces · 2 breakers. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Coastline, Dark Hickory border, hidden fasteners

T-406 · Step 7 · Rail

GOES IN

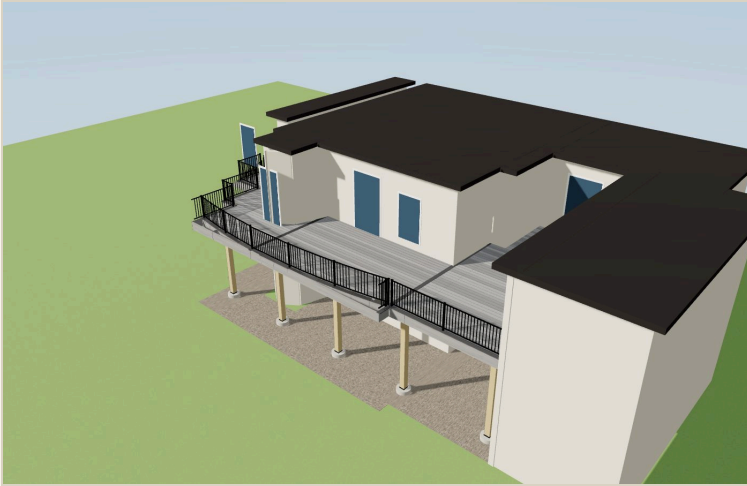
- (7) TimberTech Fulton Rail 6' x 36" level panel Black
- (6) TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)
- (4) TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt
- (8) TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt
- (5) TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt
- (1) Rust-Oleum flat black touch-up (cut ends)

HOW

- Fulton 36" — 17 posts (4 CORNER, 8 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — Fulton 36" Black



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Prices: rate card · GC at cost · 42.5% GM · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.